PORTFOLIO

BAYU AJI TRISASONGKO

Electrical Engineering Graduate | Mechanical Design, Electronics, Mechatronics & Robotics

Bringing mechanical design, electronics, and control together to turn engineering ideas into tangible prototypes. An electrical engineering graduate with a solid hands-on foundation in mechatronics, component fabrication, and rapid prototyping. Eager to contribute and learn within engineering teams developing robotics, automation, and precision mechanical systems.

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SCAN KODE QR ATAU KLIK DI MANA SAJA UNTUK BUKA PORTOFOLIO WEB
https://portobayu.vercel.app/
Bayu Aji Trisasongko · S1 Teknik Elektro, Universitas Telkom



Telkom University
S1 Teknik Elektro (2026)

Mechanical
3D CAD modeling, cycloidal gearbox design, CNC & 3D print fabrication.
`SolidWorks` `Cycloidal Reducer` `FDM 3D Print`

Electronics
Schematic design, circuit wiring, PCB layout, and power electronics.
`KiCad/Altium` `Relay Control` `Power Distribution`

Embedded Control
STM32 & Arduino MCU firmware, magnetic sensing, and closed loop.
`STM32 C/C++` `AS5600 Encoder` `PID Closed-Loop`

Prototyping
Physical mechatronics build, functional bench testing, and validation.
`Hardware Bench` `Rapid Iteration` `System Validation`

01 / PROJECTS

Engineering Work, in Practice · Selected Prototyping Showcase

Koleksi 5 proyek rekayasa fisik: Lengan Robot 5-DOF, Pendingin Surya, Robot Basket KRAI, Line Follower & Sumo Robot

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01 / FINAL PROJECT



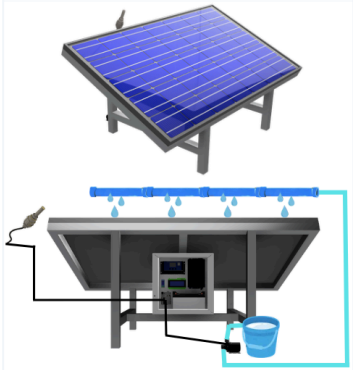
Robot Manipulator 5-DOF ↗

Prototipe lengan robot artikulasi 5-DOF bertenaga STM32F411 dengan gearbox cycloidal hasil 3D-print presisi. Mengintegrasikan encoder magnetik AS5600 closed-loop feedback untuk mereduksi backlash pada sendi artikulasi gerak, mencapai akurasi kinematika 79,8% pada beban uji 250–1.000 g.

- STM32F411 MCU
- NEMA 17 Stepper
- TB6600 Driver
- AS5600 Sensor
- TCA9548A Mux
- Cycloidal Gearb…

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02 / CAPSTONE DESIGN



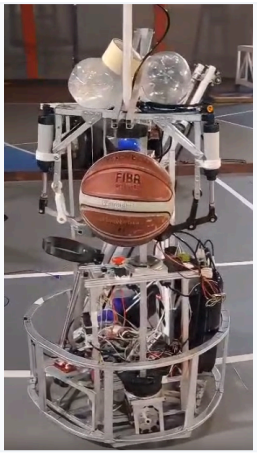
Pendingin Panel Surya Otomatis ↗

Sistem penyemprotan air otomatis untuk menurunkan temperatur operasional panel surya berbasis Arduino Uno. Memanfaatkan sensor suhu presisi dan relai pompa celup untuk memicu siklus pendinginan saat suhu >30°C dan berhenti di bawah <25°C demi memulihkan efisiensi konversi daya modul secara optimal.

- Arduino Uno R3
- RTD Temp Sensor
- 5V Relay Module
- 12V Water Pump
- 16x2 I2C LCD
- 12V Battery Pack

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03 / KONTES ROBOT ABU INDONESIA (KRAI)



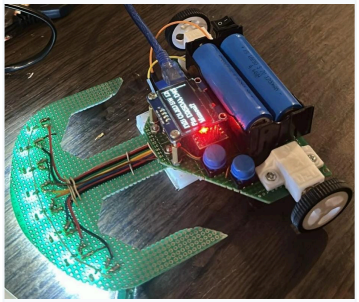
Kompetisi Robot Basket ↗

Robot penembak bola basket dinamis untuk Kontes Robot ABU Indonesia. Berperan aktif di tim mekanik merancang subsistem pelontar pneumatik 6-bar, mekanisme intake dribbler, serta chassis manuver roda omniwheel, berhasil membawa tim lolos seleksi ke Tahap Nasional ITB Jatinangor.

- 6-Bar Pneumatics
- 24V Solenoid
- RS775 DC Motor
- Omniwheel Drive
- Limit Switch & IR
- Air Regulator

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04 / PROYEK MANDIRI



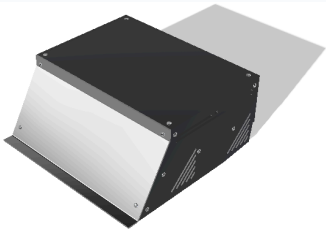
Robot Line Follower ↗

Robot pelacak garis otomatis berkecepatan tinggi dengan kemudi diferensial presisi. Dilengkapi bar sensor 8 pasang fotodioda LED, pemrosesan cepat mikrokontroler Arduino Nano, serta pengemudi motor TB6612FNG responsif dengan kalibrasi monitoring layar OLED 0.96 inci.

- Photodiode Array
- Arduino Nano
- TB6612 Driver
- N20 Gear Motor
- 0.96" I2C OLED
- 2S Li-ion Batte…

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05 / PROYEK MANDIRI



Robot Sumo ↗

Robot sumo kelas kompetisi 1 kg berdimensi 20×20 cm dengan sasis akrilik kokoh dan bilah sekop pelat aluminium. Mengandalkan 4 motor planetari PG24 torsi tinggi yang menghasilkan daya dorong 4 kg, dikendalikan nirkabel dari smartphone via Bluetooth HC-05 responsif.

- Arduino Uno
- 4x PG24 Motors
- L298N Driver
- HC-05 Bluetooth
- 3S LiPo Battery
- Acrylic Chassis

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